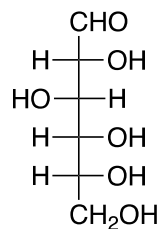


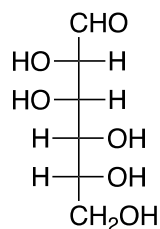
Carbohydrates

1. For each of the hexoses below, provide chair structures of the cyclic α - and β -anomers. Label each anomer clearly.

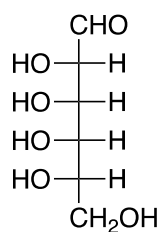
A.



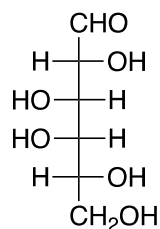
B.



C.

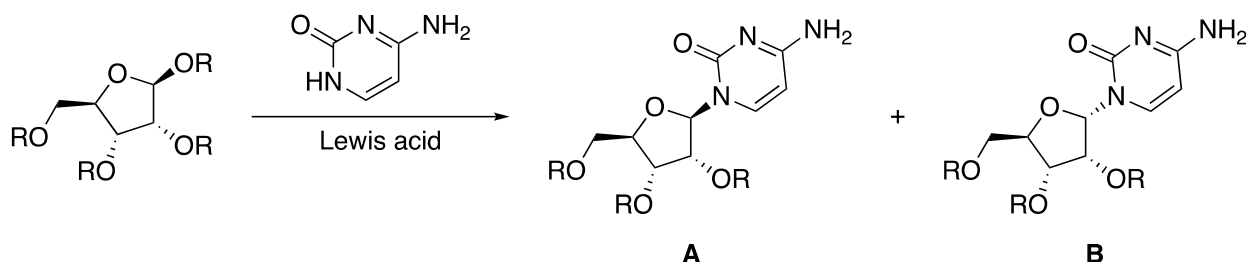


D.



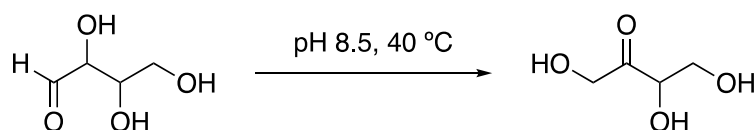
2. D-Glyceraldehyde is the simplest carbohydrate (a triose). The “D” label indicates that the stereocenter in this compound has the *S* configuration. Larger D-sugars retain this configuration.
- A. Devise a synthesis of the two possible D-tetroses from D-glyceraldehyde, striving for a mixture rather than selective formation of one or the other. Note that sodium-mercury amalgam and water reduce a carboxylic acid to an aldehyde selectively in this context.

- B. Suggest a strategy for synthesizing one of the two tetroses selectively. What reaction sets the configuration of carbon 2? How could it be modified?
3. Treatment of ribose with a Lewis acid and nitrogenous base affords nucleosides such as cytidine. An example of this reactivity is shown below.



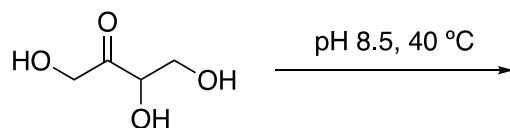
When ribose itself is used ($R = H$), a mixture of epimers **A** and **B** is formed. However, if ribose is first exhaustively acetylated ($R = Ac$), only product **A** is observed.

- A. Explain why acetylated ribose reacts in a stereoselective manner. Include the structure of at least one relevant reactive intermediate in your response.
- B. Small amounts of a constitutional isomer of **A** ($R = Ac$) are also observed in the product mixture. Draw the structure of this isomer and include a brief mechanism with curved arrows (only one or two steps are required) to illustrate its formation starting from the reactive intermediate you drew in part A.
4. The carbonyl chemistry we've studied previously in a laboratory context is highly relevant to monosaccharides. A group at Georgia Tech recently studied reactions of monosaccharides under prebiotic conditions.¹
- A. Propose a mechanism with curved arrows for the reaction below. An enolate intermediate is involved.

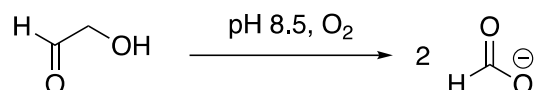


¹ Yi, R. *et al. Chem. Eur. J.* **2022**, 202202816. ([link](#))

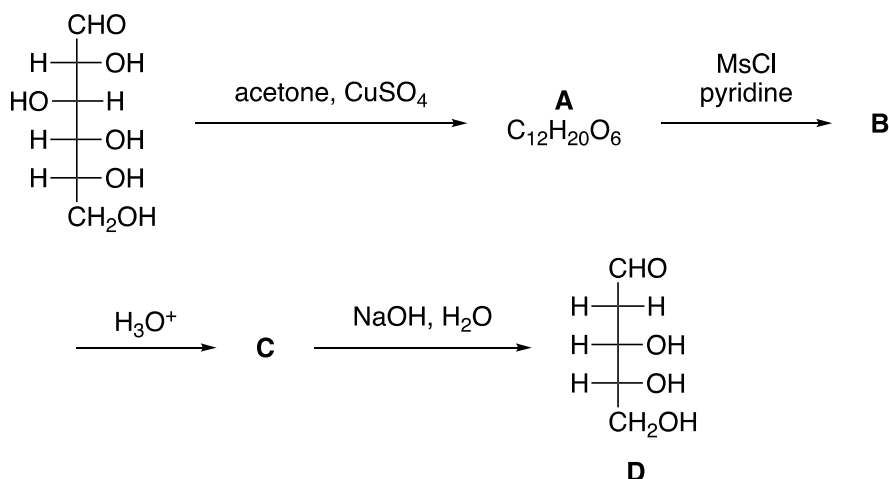
- B. When the time of the reaction above is extended, formaldehyde (H_2CO) is observed in the product mixture and the ketone above begins to disappear. Provide the structure of the other product of this fragmentation process.



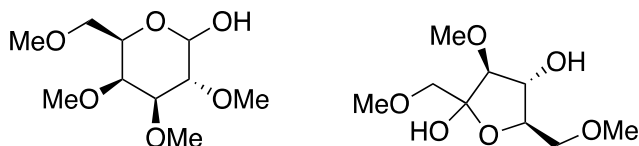
- C. Enolates can add to dioxygen (O_2) to yield α -peroxy ketones or aldehydes and related structures. Using this concept, propose a mechanism for the reaction below.



5. The sugar 2-deoxyribose is found in deoxyribonucleic acid (DNA). An elegant synthesis of this monosaccharide (**IV**) from glucose was reported in 1955. Provide the structures of compounds **A**, **B**, and **C**.

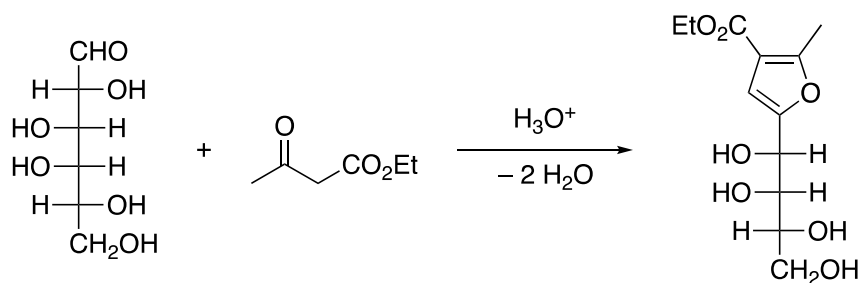


6. Treatment of an unknown disaccharide with an excess of methyl iodide followed by aqueous sulfuric acid produced the monosaccharides below. Treatment of the disaccharide in aqueous solution with 2,4-dinitrophenylhydrazine (2,4-DNP) gave a precipitate, indicating the presence of a ketone in the disaccharide.



Provide the structure of the disaccharide.

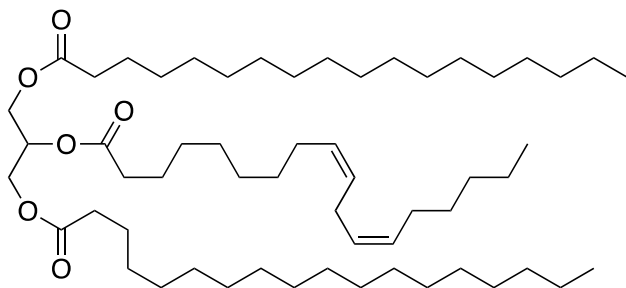
7. The Garcia Gonzalez reaction involves the condensation of a β -keto carbonyl compound with an aldose; the product is a furan. This reaction has shown promise for the conversion of sugars to value-added compounds.²



Propose a mechanism for this reaction as a list of elementary steps. While you're welcome to draw curved arrows, do not include curved arrows in your response.

Lipids

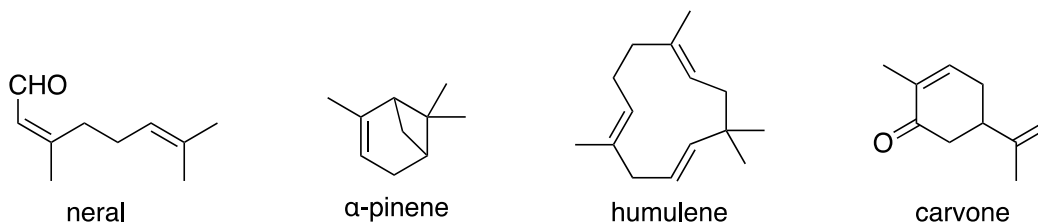
8. Draw the products formed when the triacylglycerol (triglyceride) below is treated with each of the following reagents.



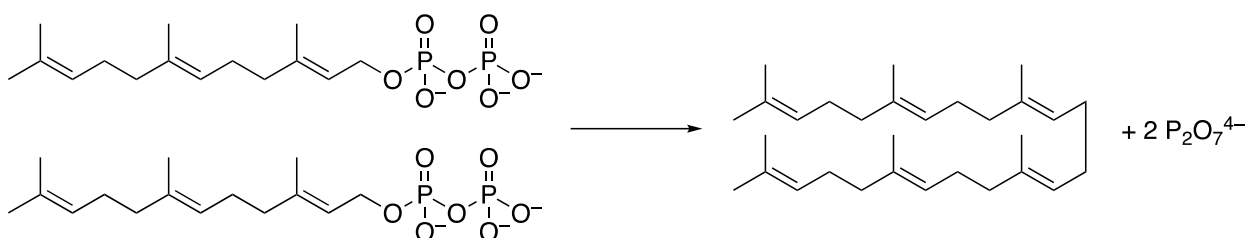
- $\text{H}_2\text{SO}_4, \text{H}_2\text{O}$
- H_2 (excess), $\text{Pd/C} \rightarrow \mathbf{B}$
- Which has a greater melting point, the starting triglyceride (let's call it **A**) or product **B**? Provide an explanation.

² Ronaghi, N. *et al. J. Org. Chem.* **2020**, 85, 15337. ([link](#))

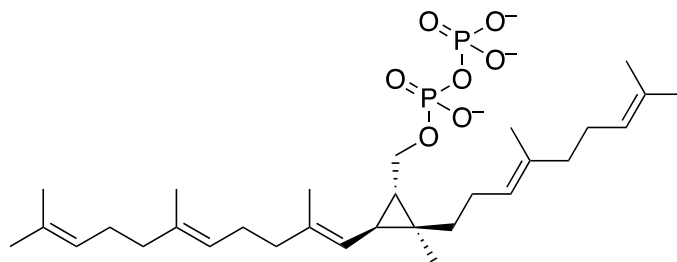
9. Each of the compounds below is biosynthesized via the introduction and elaboration of isoprene units. Circle the isoprene units in each structure.



10. In the biosynthesis of steroids, two molecules of farnesyl diphosphate react to give squalene and two molecules of pyrophosphate ($P_2O_7^{4-}$).

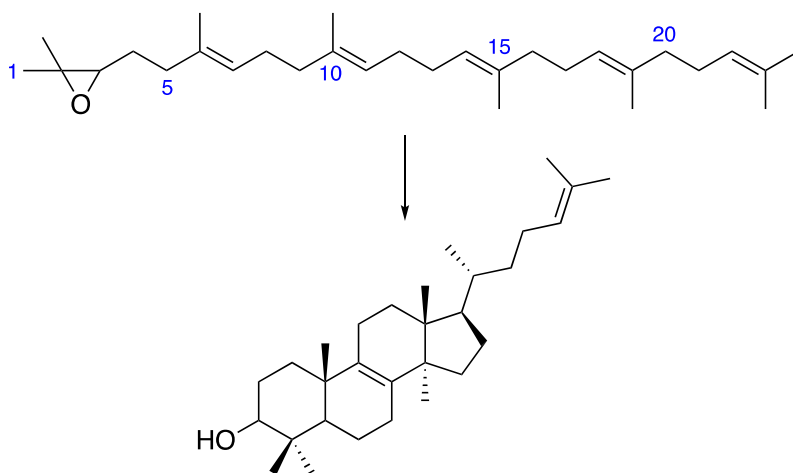


- Does this reaction amount to oxidation or reduction of farnesyl diphosphate?
- Considering your response to the last problem, identify a cofactor that would be needed to promote this reaction. What is the oxidizing or reducing agent?
- The mechanism of this reaction is remarkable—the molecule below has been identified as a reactive intermediate.

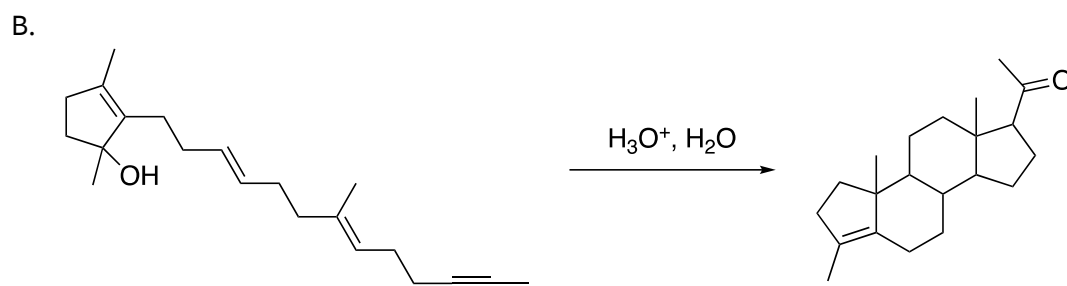
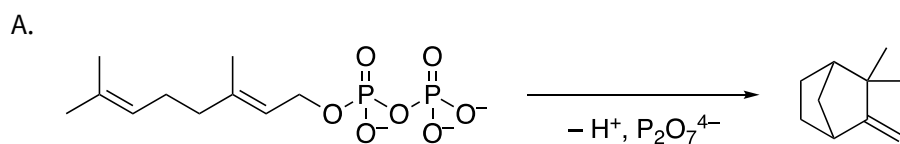


Propose a mechanism for the conversion of the molecule above to squalene and one molecule of pyrophosphate. Include curved arrows in your response (unreactive portions of the molecule can be abbreviated as “R”).

11. The core of a wide variety of steroids is formed via cyclization of squalene-2,3-epoxide. Incredibly, this “zipping up” occurs in a single elementary step in the active site of the enzyme lanosterol synthase.



- A. Use the given numbers in squalene-2,3-epoxide to label the corresponding carbons in lanosterol.
- B. Using a generic base “B” and acid “HA” to represent acidic and basic groups in the enzyme, provide a single-step mechanism with curved arrows for this transformation. *Hint:* it involves several A_E electron flows “capped” by proton transfers, all occurring at once.
12. For each of the following reactions, propose a mechanism as a series of elementary step labels. While you’re welcome to draw curved arrows to think through these, submit only the sequence of elementary steps in your response.



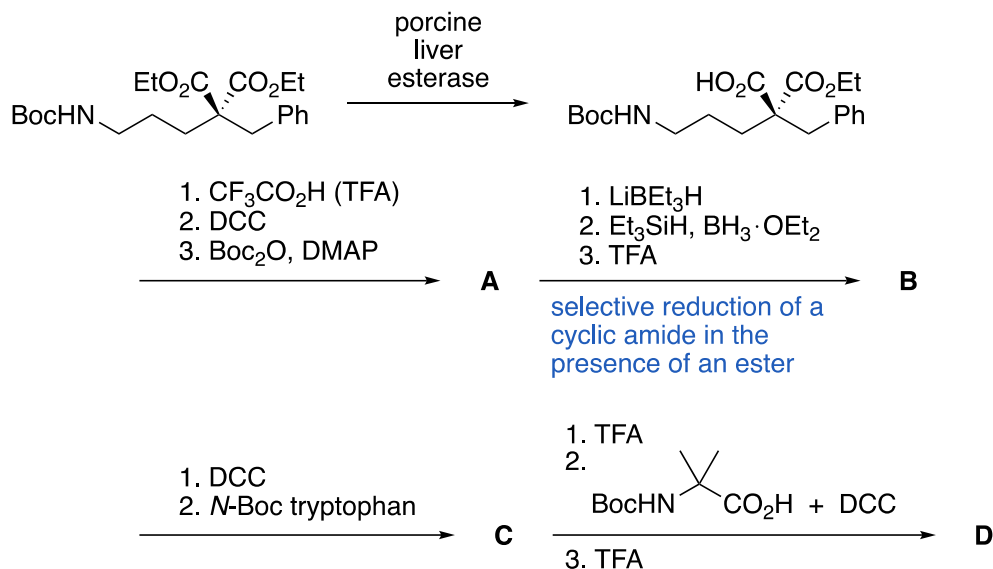
Amino Acids; Polypeptide Synthesis and Characterization

13. Complete the table below by drawing the amino acids in the rows in their dominant ionization states at the pH values in each column. Acidity constants for the amino acids can be found [here](#).

	pH 1.0	pH 5.0	pH 8.0	pH 13.0
aspartic acid				
lysine				
histidine				
arginine				

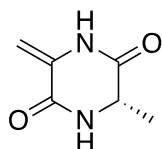
14. Draw the structure of a polypeptide with the sequence *N*-Thr-Val-Pro-C at pH 7.0. This tripeptide is found in the chemotherapeutic agent [dactinomycin](#).

15. Growth hormone secretagogues are small molecules that induce the release of growth hormones in humans. One such compound was [synthesized](#) using the sequence below. Fill in the structures of synthetic intermediates **A** – **D**. DMAP = *N,N*-dimethylaminopyridine, a catalyst for the acylation of nucleophiles.

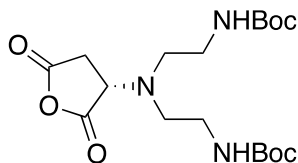


16. Devise synthesis of the following compounds from one or more of the natural amino acids. Other sources of carbon can also be used.

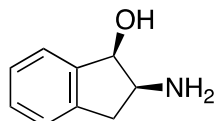
A.



B.



C.



D.

